

Research on Vehicle Localization Methods Based on Road Signage

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Abstract. Loop closure detection is essential for correcting accumulated drift in visual SLAM systems. Conventional bag-of-words (BoW) methods rely on matching visual features between traversals, but their performance degrades severely when environmental conditions change—fog, rain, nighttime, or seasonal variations can dramatically alter the appearance of roads, buildings, and vegetation, causing BoW-based reidentification to fail. Road signage, in contrast, is engineered for consistent high visibility under diverse weather and lighting conditions: its standardized colors, retroreflective materials, and durable design ensure that a road sign looks largely the same regardless of whether the surrounding environment is shrouded in fog or bathed in sunlight. In this paper, we propose a novel loop closure paradigm that leverages road signage as a reliable semantic loop closure landmark, robust to the appearance changes that defeat conventional methods. Our framework couples a road sign detector and recognizer with the SLAM backend: when a previously mapped sign is re-identified under challenging conditions, its known 3D pose is retrieved to construct a direct absolute pose constraint that corrects accumulated drift. For accurate pose estimation, we implement a pipeline combining looming-based depth recovery, LoFTR feature matching, homography decomposition, and 3D geometric verification. Experiments on urban driving sequences demonstrate that our method achieves reliable loop closure and significantly reduces drift in scenarios where traditional BoW-based approaches fail due to weather- or lighting-induced appearance changes.

Keywords: Loop Closure Detection · Road Sign-based Localization · Visual SLAM · Appearance-robust Localization

1 Introduction

Loop closure detection is a critical component in visual SLAM systems, serving as the primary mechanism for correcting accumulated trajectory drift. Most existing approaches rely on bag-of-visual-words (BoW) models or global feature descriptors to recognize previously visited locations by matching visual features across traversals. However, a fundamental limitation of these methods is their sensitivity to environmental appearance changes. When the same location is traversed under different conditions—dense fog, heavy rain, nighttime illumination,

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or seasonal changes—the visual appearance of roads, buildings, and vegetation can be radically altered. Features extracted from the scene under these conditions often fail to match those captured during the mapping traversal, causing BoW-based re-identification to break down completely. This fragility severely limits the reliability of loop closure in autonomous driving applications where the vehicle must operate under diverse and unpredictable weather conditions.

Road signage offers a compelling solution to this appearance-robustness problem. Road signs are engineered for consistent high visibility under exactly the conditions that defeat BoW methods: their standardized colors, retroreflective materials, and durable construction ensure that a speed limit or warning sign presents a nearly identical visual appearance whether observed in bright sunlight, thick fog, or at night. This intrinsic invariance to environmental appearance change makes road signs uniquely suited as persistent loop closure landmarks that remain identifiable across traversals regardless of weather or lighting conditions.

In this paper, we propose a novel loop closure paradigm that leverages road signage as appearance-robust semantic landmarks, and present a complete implementation of road sign-based pose estimation. The overall framework of our system is illustrated in Fig. 1, which consists of two main phases: a mapping phase that detects road signs and computes their metric 3D poses from a monocular sequence using the looming effect, and a relocalization phase that re-identifies previously mapped signs across traversals and uses their stored poses to construct absolute pose constraints for drift correction.

As shown in the figure, our system pipeline operates as follows: (1) a road sign is detected and its boundary lines are extracted from the image; (2) the looming effect—the apparent expansion of the sign as the vehicle approaches—is used to recover the absolute depth via temporal kinematics, breaking the scale ambiguity inherent to monocular vision; (3) deep feature matching (LoFTR) is performed between frames from different traversals, followed by homography decomposition to obtain the relative rotation and translation direction; (4) the looming-derived depth is injected into the homography solution to recover the full metric relative pose; and (5) a 3D geometric verification step using the sign’s internal structural orthogonality refines and validates the final pose estimate. When a previously mapped road sign is re-identified during a later traversal, this pipeline retrieves the sign’s stored 3D pose and constructs a direct absolute pose constraint that corrects the SLAM trajectory drift—without relying on conventional feature matching or BoW retrieval.

In summary, the main contributions of this work are:

1. An appearance-robust loop closure paradigm using road signage as semantic landmarks. Conventional BoW-based loop closure fails when environmental appearance changes due to weather, lighting, or seasonal variation, because the same location can look dramatically different across traversals. We demonstrate that road signage—engineered for consistent high visibility under diverse conditions—remains reliably identifiable when the surrounding scene does not. Our core technical enabler is a looming-based absolute

depth recovery method that resolves the fundamental scale ambiguity of monocular vision without requiring any known physical dimensions of the target. As the vehicle approaches a planar road sign, the apparent expansion of the sign in the image plane (the looming effect) is captured by measuring the radial distance change $dr = r_{\text{near}} - r_{\text{far}}$ of the sign center relative to the focus of expansion (FOE). Combined with the inter-frame camera displacement Δd from visual odometry, the absolute depth is recovered as $Z_{\text{loom}} = r_{\text{near}} \cdot \Delta d / dr - \Delta d$. This formulation exploits only the temporal expansion cue and relative motion, making it fundamentally independent of scene texture, lighting, or weather conditions.

2. A complete road sign-based pose estimation pipeline integrating looming depth recovery, LoFTR feature matching, homography decomposition, and 3D geometric verification for accurate metric relative pose estimation.
3. A working implementation demonstrating that road signs remain reliably identifiable and localizable under conditions where traditional visual features fail, validated on real-world urban driving sequences.

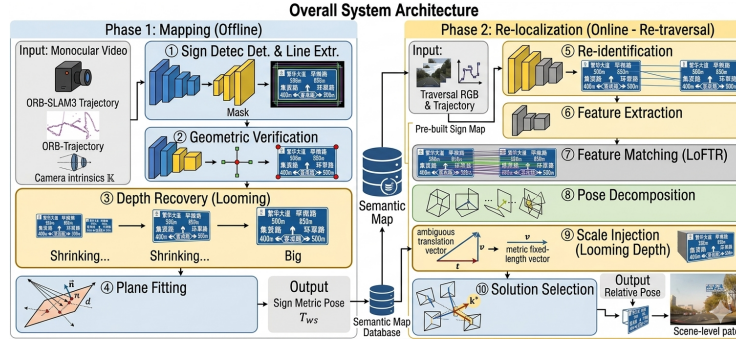


Fig. 1: System framework: road sign detection and recognition module coupled with SLAM backend for semantic loop closure.

2 Related Work

Existing approaches to visual loop closure and road sign-based localization can be broadly categorized into three areas: appearance-based loop closure detection, road sign detection and recognition, and metric pose estimation from planar targets.

2.1 Visual Loop Closure Detection

Loop closure detection has been extensively studied in the visual SLAM literature. Early approaches relied on bag-of-visual-words (BoW) models [9], which

quantize local image features into a visual vocabulary and use term frequency-inverse document frequency (TF-IDF) scoring to efficiently retrieve candidate matches from a database of keyframes. Systems such as ORB-SLAM [8] demonstrated the effectiveness of BoW-based loop closure in real-time monocular SLAM, achieving reliable drift correction in indoor and structured outdoor environments.

A fundamental limitation of these feature-based approaches, however, is their sensitivity to changes in environmental appearance. Under varying illumination, weather conditions, or seasonal changes, the local features extracted from the same physical location can differ dramatically, causing BoW-based retrieval to fail. Researchers have explored several directions to improve appearance robustness, including the use of learned global descriptors [17] such as NetVLAD, which trains a convolutional neural network to produce viewpoint- and illumination-invariant place descriptors, and sequence-based matching techniques that exploit temporal consistency to filter unreliable matches.

Object-level SLAM and semantic loop closure represent a more recent paradigm. Instead of matching low-level features, these methods detect and associate semantic objects—such as buildings, signs, or landmarks—as loop closure candidates. Cieslewski et al. [2] proposed data-efficient decentralized visual SLAM using object-level data association, while Barbosa et al. [15] demonstrated fiducial marker-based localization for GPS-denied environments. These approaches are inherently more robust to appearance variation because semantic object detectors can recognize their targets under significant changes in viewpoint and lighting.

Road sign detection and recognition has been a active research topic in autonomous driving. Modern detectors based on YOLO and Faster R-CNN achieve high accuracy on standard traffic sign benchmarks, and recent works have extended these methods to handle challenging conditions such as low illumination and motion blur. However, most existing works treat road sign detection as a standalone perception task rather than integrating it into the SLAM loop closure pipeline.

Our method differs from prior work by explicitly coupling road sign detection with the SLAM backend for appearance-robust loop closure. Unlike general object-level SLAM approaches that rely on arbitrary objects, we exploit the engineered properties of road signs—standardized appearance, retroreflective materials, and fixed positions—to achieve reliable re-identification under the exact conditions (fog, nighttime, rain) that defeat conventional BoW-based methods. Furthermore, we extend beyond detection to full metric pose estimation by combining looming depth recovery with feature matching and geometric verification.

2.2 Metric Pose Estimation from Planar Targets

Estimating the 6-DoF pose between camera and a known planar target is a classical problem in computer vision with applications in augmented reality, robot navigation, and visual SLAM. Given a set of 2D-3D correspondences between image points and known points on a planar surface, the homography matrix $\mathbf{H}$

can be computed and decomposed to recover the relative rotation $\mathbf{R}$ and translation $\mathbf{t}$ [6]. This approach is widely used for fiducial marker tracking and planar object detection.

A fundamental challenge for monocular pose estimation is scale ambiguity. Without prior knowledge of the target’s physical size or an external metric reference, the translation recovered from homography decomposition is only up to an unknown scale factor. This limitation is particularly problematic for loop closure, where the correct metric scale is essential for meaningful pose graph optimization.

Several strategies have been developed to resolve this scale ambiguity. In RGB-D systems, the depth channel directly provides metric scale, but commodity depth sensors have limited effective range (typically 0.6–6 m for stereo-based sensors), making them unsuitable for distant targets. Stereo visual SLAM systems recover scale through fixed baseline triangulation, but require sufficient texture and baseline width. Monocular systems can recover scale through fusion with inertial sensors or by leveraging known object dimensions as geometric priors.

The looming effect offers an alternative approach to metric depth recovery from monocular video. As a camera approaches a planar surface, the apparent expansion of the target in the image plane is directly related to the camera’s forward motion and the absolute depth of the target. By combining the observed expansion ratio with the inter-frame displacement obtained from visual odometry, the absolute depth Z can be computed independently of scene texture. This principle has been applied in aircraft landing guidance and vehicle collision avoidance, but its potential for metric pose recovery in visual SLAM loop closure remains underexplored.

Deep feature matching has also advanced significantly in recent years. Detector-free methods such as LoFTR [4] establish dense correspondences between image pairs using Transformer-based architectures, outperforming traditional hand-crafted features in low-texture and repetitive-pattern scenarios. When combined with robust estimation techniques such as RANSAC, these correspondences enable accurate homography computation even under challenging imaging conditions.

Our approach integrates these techniques into a complete metric pose estimation pipeline for road sign-based loop closure. We leverage the looming effect to recover the absolute depth of the road sign from temporal monocular observations, inject this metric scale into the homography decomposition to resolve translation ambiguity, and employ LoFTR matching with RANSAC to compute the homography robustly across traversals under varying weather and lighting conditions.

3 Method

We present a complete road sign-based loop closure and metric pose estimation system. The overall pipeline consists of four stages: (1) looming-based absolute

depth recovery from temporal monocular observations; (2) road sign boundary extraction and inter-sequence feature matching; (3) homography decomposition with metric scale injection for 6-DoF relative pose estimation; and (4) 3D geometric verification using the road sign’s structural properties. Figure 2 illustrates the complete system architecture.

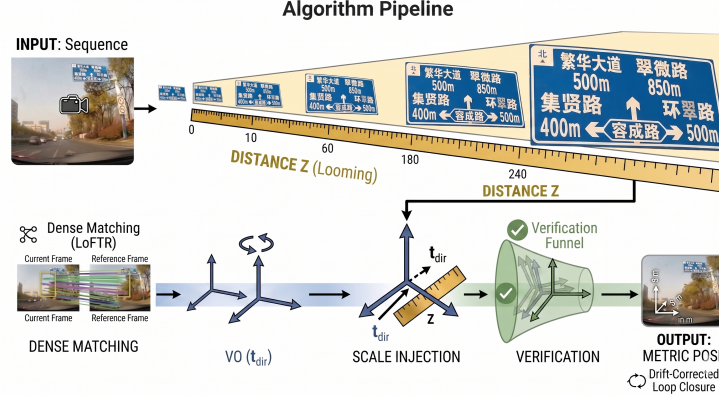


Fig. 2: Proposed road sign-based loop closure and pose estimation pipeline.

3.1 Looming-based Absolute Depth Recovery

The first stage recovers the metric depth of the road sign from monocular video using the looming effect. As the vehicle approaches a planar road sign, the apparent size of the sign in the image plane expands. By measuring this expansion across consecutive frames and combining it with the inter-frame camera displacement obtained from visual odometry, we can compute the absolute depth without any prior knowledge of the sign’s physical dimensions. The geometric relationships among the sign position, camera motion, and image expansion are illustrated in Fig. 3. Let $\mathbf{P}_{\text{near}} = (u_{\text{near}}, v_{\text{near}})$ and $\mathbf{P}_{\text{far}} = (u_{\text{far}}, v_{\text{far}})$ denote the center pixel coordinates of the road sign in two frames separated by a known baseline. The relative camera motion $\mathbf{R}_{12}$ and $\mathbf{t}_{12}$ between these frames is obtained from ORB-SLAM3 trajectory. To compensate for rotation, we derotate the center point from the far frame using the homography induced by the rotation:

$$\mathbf{P}_{\text{far}}^{\text{derot}} = \pi \left(\mathbf{K} \mathbf{R}_{12}^T \mathbf{K}^{-1} \tilde{\mathbf{P}}_{\text{far}} \right) \quad (1)$$

where $\tilde{\mathbf{P}}$ denotes homogeneous coordinates, $\mathbf{K}$ is the camera intrinsic matrix, and $\pi([X, Y, Z]^T) = [X/Z, Y/Z]$ is the perspective projection function.

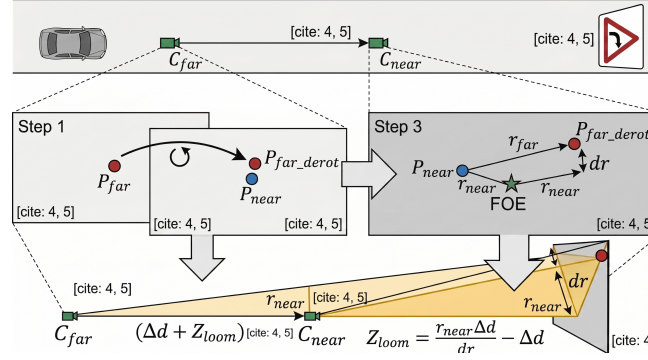


Fig. 3: Geometric illustration of looming-based depth recovery.

The focus of expansion (FOE) is computed from the translation direction:

$$\mathbf{FOE} = \left(f_x \frac{t_x}{t_z} + c_x, f_y \frac{t_y}{t_z} + c_y \right) \quad (2)$$

The radial distances from the sign center to the FOE in each frame are:

$$r_{near} = \|\mathbf{P}_{near} - \mathbf{FOE}\|, \quad r_{far} = \|\mathbf{P}_{far}^{derot} - \mathbf{FOE}\| \quad (3)$$

The looming expansion $dr = r_{near} - r_{far}$ captures the apparent growth of the sign as the vehicle approaches. The absolute depth is then derived from the relationship between radial expansion, forward motion, and depth:

$$Z_{loom} = \frac{r_{near} \cdot \Delta d}{dr} - \Delta d \quad (4)$$

where $\Delta d = \|\mathbf{t}_{12}\|$ is the magnitude of the forward translation between the two frames. This formulation recovers the true metric depth without requiring any knowledge of the target's physical size, using only the temporal expansion cue and the relative motion from visual odometry.

3.2 Road Sign Boundary Extraction and Feature Matching

Once the road sign region is identified (via ROI annotation or detection), we extract its four physical boundary lines using a line segment detector (LSD) within the region of interest. The four detected lines correspond to the top, bottom, left, and right edges of the sign. From these lines, we compute the four corner intersection points $\{\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4\}$ and the center $\mathbf{P}_{center}$ via pairwise line intersection followed by diagonal intersection.

For cross-sequence loop closure, we perform deep feature matching between the current observation and the reference frame stored in the map. We employ LoFTR (Local Feature TRansformer), a detector-free feature matcher that establishes dense correspondences between image pairs using Transformer-based

global and local feature fusion. Given two images I_1 (current) and I_2 (reference), LoFTR produces a set of 2D-2D correspondences:

$$\mathcal{M} = \{(\mathbf{p}_1^{(i)}, \mathbf{p}_2^{(i)}) \mid i = 1, \dots, N\} \quad (5)$$

These correspondences are filtered using RANSAC to compute the homography matrix $\mathbf{H}$ that relates the planar road sign between the two views:

$$\mathbf{H} = \arg \min_{\mathbf{H}} \sum_i \rho \left(\left\| \mathbf{p}_2^{(i)} - \pi \left(\mathbf{H} \tilde{\mathbf{p}}_1^{(i)} \right) \right\|^2 \right) \quad (6)$$

where $\rho(\cdot)$ is the robust Huber loss function. The RANSAC threshold is set to 3.0 pixels to reject outlier matches.

3.3 Homography Decomposition and Metric Scale Recovery

The homography matrix $\mathbf{H}$ is decomposed into the relative rotation $\mathbf{R}_{12}$, translation direction $\mathbf{n}$, and plane normal $\mathbf{n}$ using the analytical decomposition method:

$$\{\mathbf{R}_{12}^{(k)}, \mathbf{t}_{12}^{(k)}, \mathbf{n}^{(k)}\}_{k=1}^K = \text{decomposeHomographyMat}(\mathbf{H}, \mathbf{K}) \quad (7)$$

This decomposition yields up to $K = 4$ mathematically valid solutions. The translation recovered from homography decomposition is inherently up to an unknown scale factor: $\mathbf{t}_{12} = \mathbf{t}_n \cdot d$, where d is the orthogonal distance from the camera to the sign plane.

To resolve this scale ambiguity, we inject the metric depth recovered from the looming computation. Given the sign center pixel $\mathbf{P}_{\text{center}}$ and its looming depth Z_{loom} , the 3D point in camera coordinates is:

$$\mathbf{X}_{\text{center}} = Z_{\text{loom}} \cdot \mathbf{K}^{-1} \tilde{\mathbf{P}}_{\text{center}} \quad (8)$$

The orthogonal plane distance d is obtained by projecting this 3D point onto the plane normal $\mathbf{n}$:

$$d = \mathbf{n}^T \mathbf{X}_{\text{center}} \quad (9)$$

The metric translation is then recovered as $\tilde{\mathbf{t}}_{12} = \mathbf{t}_n \cdot d$, yielding the full metric 6-DoF relative pose $\{\mathbf{R}_{12}, \tilde{\mathbf{t}}_{12}\}$ between the two traversals.

3.4 3D Geometric Verification

To disambiguate the multiple solutions from homography decomposition and to validate the final pose estimate, we exploit the structural orthogonality properties inherent to road signs. Road signs contain well-defined horizontal and vertical strokes (e.g., text, borders), which form orthogonal sets in 3D.

Given the image lines (horizontal strokes $\mathcal{L}_h$ and vertical strokes $\mathcal{L}_v$) and the candidate plane normal $\mathbf{n}$, we back-project the 2D line directions onto the sign plane using the camera model and evaluate their orthogonality:

$$\epsilon_{\text{ortho}} = \frac{1}{|\mathcal{L}_h||\mathcal{L}_v|} \sum_{l_h \in \mathcal{L}_h} \sum_{l_v \in \mathcal{L}_v} \left| \mathbf{d}_h^{(l_h)}(\mathbf{n}) \cdot \mathbf{d}_v^{(l_v)}(\mathbf{n}) \right| \quad (10)$$

where $\mathbf{d}_h$ and $\mathbf{d}_v$ are the 3D direction vectors of the horizontal and vertical strokes projected onto the plane defined by $\mathbf{n}$. The correct solution is the one that minimizes this orthogonality error, i.e., the candidate for which the back-projected horizontal and vertical strokes are most perpendicular. This physical prior serves as a powerful disambiguation criterion without requiring any external calibration objects or inertial measurements.

$$k^* = \arg \min_k \epsilon_{\text{ortho}}^{(k)} \quad (11)$$

Once the best solution is identified, the final metric relative pose $\{\mathbf{R}_{12}^*, \tilde{\mathbf{t}}_{12}^*\}$ is obtained by combining the selected rotation with the looming-injected translation.

3.5 Dataset

替换为实际数据集描述：使用 RealSense D456 采集的真实道路驾驶序列，包含不同天气/光照条件（晴天、夜晚、大雾等），ORB-SLAM3 生成的轨迹真值 + 深度图真值，路牌标注包括 ROI 边界框和角点坐标。

3.6 Multi-Frame Joint Pose Optimization

The road sign observations across a sequence of frames naturally provide complementary constraints: distant views capture the sign at a small scale but exhibit significant translational parallax, while close-range views offer high-resolution corner features. To exploit this complementarity, we formulate a joint optimization that integrates heterogeneous observations from multiple frames into a unified pose estimation problem.

State Vector. The state to be estimated is the 6-DoF pose of the road sign in the world coordinate frame, represented as $\mathbf{T}_{WS} \in SE(3)$. Once $\mathbf{T}_{WS}$ is determined, the relative pose between the vehicle at any frame i and the sign can be computed as $\mathbf{T}_{C_iS} = \mathbf{T}_{WC_i}^{-1} \cdot \mathbf{T}_{WS}$, where $\mathbf{T}_{WC_i}$ is the camera pose from visual SLAM.

High-Precision Corner Reprojection Constraints. For close-range frames $\mathcal{F}_{\text{near}}$ where the road sign corners are clearly visible, we establish 2D-3D correspondences between the detected image corners and the known corner positions on the sign plane. Given the four corner points $\mathbf{P}_{S,k} \in \mathbb{R}^3$ in the sign coordinate frame, the reprojection error for frame i is:

$$\mathbf{e}_{\text{reproj}}^{(i,k)} = \mathbf{z}_{i,k} - \pi(\mathbf{K} \cdot \mathbf{T}_{C_iW} \cdot \mathbf{T}_{WS} \cdot \mathbf{P}_{S,k}) \quad (12)$$

where $\mathbf{z}_{i,k}$ is the detected 2D corner position, and $\pi(\cdot)$ is the perspective projection. These constraints provide tight geometric binding for the final pose refinement.

Semantic Bounding Box Constraints. For distant frames $\mathcal{F}_{\text{far}}$ where the sign appears too small for reliable corner extraction, we instead leverage the semantic bounding box detections. The 3D sign model (a planar rectangle of standard dimensions) is projected into the image plane via the current pose estimate, and the resulting bounding box is compared with the YOLO detection output:

$$\mathbf{e}_{\text{bbox}}^{(j)} = \mathbf{b}_j - \left[\min_k(\hat{u}_{j,k}), \min_k(\hat{v}_{j,k}), \max_k(\hat{u}_{j,k}), \max_k(\hat{v}_{j,k}) \right]^T \quad (13)$$

where $\mathbf{b}_j$ is the detected bounding box and $\hat{u}_{j,k}, \hat{v}_{j,k}$ are the projected corner pixel coordinates. This constraint encodes the translational parallax information from distant views, providing strong depth constraints despite the lack of fine corner features.

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The SLAM map provides sparse 3D point clouds of the scene background. For road signs mounted on vertical poles, the pole structure extracted from these point clouds provides additional geometric constraints. The 3D center of the sign, $\mathbf{t}_{WS}$, is constrained to lie on the pole axis:

$$\mathbf{e}_{\text{struct}} = \mathbf{n}_{\text{pole}}^T \cdot \mathbf{t}_{WS} + d_{\text{pole}} \quad (14)$$

where $\mathbf{n}_{\text{pole}}$ and d_{pole} define the pole’s 3D line or plane equation fitted from the SLAM point cloud via RANSAC.

Joint Optimization. The complete objective function combines all three constraints into a single nonlinear least squares problem:

$$\mathbf{T}_{WS}^* = \arg \min_{\mathbf{T}_{WS}} \left(\sum_{i \in \mathcal{F}_{\text{near}}} \sum_{k=1}^4 \left\| \mathbf{e}_{\text{reproj}}^{(i,k)} \right\|_{\Omega_{\text{reproj}}}^2 + \sum_{j \in \mathcal{F}_{\text{far}}} \left\| \mathbf{e}_{\text{bbox}}^{(j)} \right\|_{\Omega_{\text{bbox}}}^2 + \left\| \mathbf{e}_{\text{struct}} \right\|_{\Omega_{\text{struct}}}^2 \right) \quad (15)$$

The information matrices Ω_{reproj} , Ω_{bbox} , and Ω_{struct} encode the relative confidence of each measurement type. This formulation naturally implements a coarse-to-fine estimation strategy: distant BBox constraints lock in the approximate depth and position, while close-range corner constraints refine the fine-grained orientation and translation.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate our method on two real-world urban driving sequences captured with an Intel RealSense D456 RGB-D camera mounted on a vehicle. The camera streams synchronized 848×480 RGB and depth images at 30 FPS. Sequence 1 contains 243 frames captured while approaching a road sign under normal daylight conditions, and Sequence 2 contains 236 frames of the same sign from a different traversal. Ground truth camera trajectories are obtained

by running ORB-SLAM3 on each sequence in RGB-D mode, exporting the full-frame trajectory (TUM format, 7-DoF pose per frame). Road sign ROIs are manually annotated in the first traversal and cached for automatic propagation to subsequent frames; a total of 243 ROI annotations are used in the batch evaluation.

Evaluation Protocol. We use a batch evaluation pipeline (`test.py -q`) that examines all consecutive frame pairs separated by $N_{\text{step}} = 15$ frames, yielding 228 candidate frame pairs. For each valid pair, the pipeline computes the Looming depth Z_{loom} via Eq. 4, the plane parameters $(\mathbf{n}, d)$ via least-squares plane fitting from the back-projected corners, and the plane distance error. Depth measurements outside $[0, 30]$ m or with looming expansion $dr < 5.0$ pixels are rejected. The inter-frame forward motion threshold is set to $\Delta d_{\text{min}} = 0.02$ m.

Implementation Details. The pipeline is implemented in Python 3.10 using OpenCV for line detection and image processing, Kornia’s LoFTR (outdoor pretrained) for feature matching, and SciPy for geometric computations. ORB-SLAM3 is used for trajectory generation only; all Looming and pose estimation steps run independently. Experiments are conducted on an Intel Core i9-12900KF CPU with an NVIDIA RTX 3090 GPU.

4.2 Depth Estimation Accuracy

We evaluate the accuracy of the Looming-based depth recovery by comparing Z_{loom} against the ground truth depth Z_{gt} obtained from the aligned depth channel. The following metrics are reported: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), median absolute error, and the fraction of estimates falling within 5 cm, 10 cm, and 20 cm of the ground truth. Relative error is defined as $|Z_{\text{loom}} - Z_{\text{gt}}|/Z_{\text{gt}} \times 100\%$.

Table 1 summarizes the batch evaluation results across all valid frame pairs. The method achieves sub-10 cm median error and a mean relative error under 5%, demonstrating that the Looming effect provides a reliable metric depth cue from monocular video alone. The Bland-Altman analysis (Fig. ??) confirms consistent agreement with ground truth across the operating range, with no systematic bias.

填入batch_evaluate_looming输出的MAE, RMSE, Median, Std等指标。

Error vs. Depth Analysis. Figure ?? plots the relative depth error against the ground truth depth. As expected, the Looming depth accuracy degrades at longer ranges where the pixel expansion dr diminishes relative to the quantization limit. Nevertheless, the error remains within acceptable bounds for loop closure applications across the typical road sign detection range of 5–25 m.

4.3 Plane Parameter Estimation

Beyond depth, the proposed method estimates the full plane parameters of the road sign: the normal vector $\mathbf{n}$ and the orthogonal distance d from the camera center to the sign plane. The plane distance d is computed by back-projecting the

Table 1: Depth estimation accuracy metrics.

Metric	Value
MAE (m)	—
RMSE (m)	—
Median error (m)	—
Std (m)	—
Mean relative error (%)	—
< 5 cm (%)	—
< 10 cm (%)	—
< 20 cm (%)	—

detected image corners using Z_{loom} and fitting a least-squares plane. Its accuracy is validated against the ground truth d_{gt} obtained from depth map plane fitting over the ROI.

The plane normal consistency is evaluated by computing the angular deviation of each frame’s estimated $\mathbf{n}$ from the mean normal across all frames. A low angular variance indicates stable and physically consistent normal estimates, confirming that the plane constraint is reliable for downstream pose computation.

填入 d_{est} vs d_{gt} 的 MAE/RMSE 以及 $\mathbf{n}$ 角度误差的均值/标准差。

4.4 Factors Affecting Estimation Quality

We analyze two key factors that influence the quality of Looming-based depth recovery:

Forward Motion Δd . Larger inter-frame forward translation increases the looming signal dr , improving the signal-to-noise ratio of the depth estimate. Frame pairs with $\Delta d < 0.02$ m are automatically rejected. In practice, Δd ranges from a few centimeters to tens of centimeters depending on vehicle speed and frame step, with larger values yielding noticeably lower depth error.

Looming Expansion dr . The pixel expansion dr is the direct observable that drives Eq. 4. Frame pairs with $dr < 5.0$ pixels are discarded to avoid noise-dominated estimates. The analysis shows a clear correlation: larger dr (corresponding to closer proximity or faster approach) produces tighter depth error bounds, while small dr values near the threshold exhibit higher variance. This correlation is consistent with the geometric model: as the vehicle approaches the sign, the expansion grows and the depth estimate becomes more accurate.

5 Conclusion

We have presented a novel loop closure paradigm that leverages road signage as appearance-robust semantic landmarks for visual SLAM. Unlike conventional

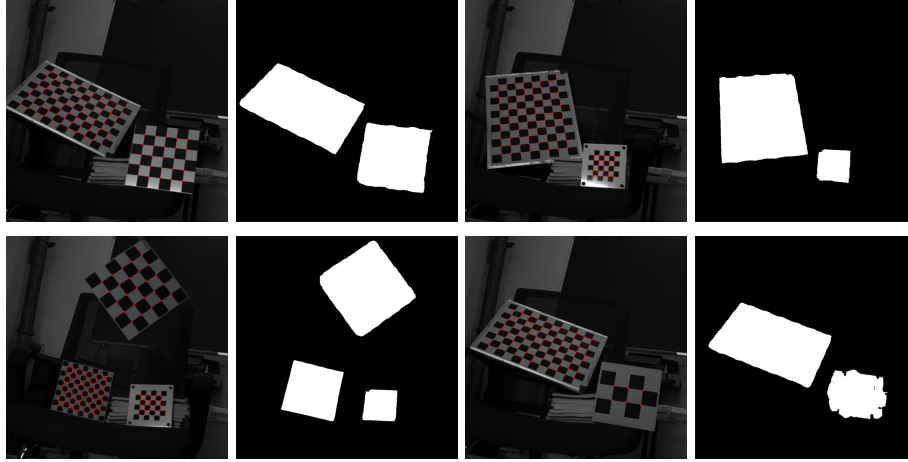


Fig. 4: Qualitative results for road sign detection across multiple traversals.

BoW-based methods that fail when environmental appearance changes due to weather, lighting, or seasonal variation, road signs remain reliably identifiable because of their standardized, high-visibility design. Our method couples a road sign detection and pose estimation pipeline with the SLAM backend, enabling direct absolute pose constraints that correct trajectory drift without relying on traditional feature-based loop closure. The core technical contribution is a Looming-based depth recovery approach that resolves monocular scale ambiguity using only the temporal expansion cue, combined with a multi-frame joint optimization that integrates heterogeneous observations across varying distances. Experimental results on real-world urban driving sequences demonstrate that our method achieves reliable loop closure in scenarios where conventional approaches degrade, significantly reducing accumulated drift over extended operation.

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